

Tutorials for the course BIE-MA1

Antonella Marchesiello, Ph.D.

Faculty of Information Technology
Czech Technical University

February 23, 2026

Tutorial n. 2

Sequences

sequences: basic properties; “Big O and little o ”; monotonic sequences

Notation

$(a_n)_{n=1}^{\infty}, (a_n)$ sequence
 $U_a(\epsilon)$ neighbourhood of size ϵ of a point $a \in \mathbb{R}$
 $\overline{\mathbb{R}}$ extended real line $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$
 $\mathcal{O}, o$ Big and little o

2.1 Asymptotic properties of sequences: “Big O and little o ”

Exercise 2.1: Show that

- a) $3n^2 + 2n^4 = \mathcal{O}(n^4)$ for $n \rightarrow +\infty$,
- b) $(-1)^n n = \mathcal{O}(n)$ for $n \rightarrow +\infty$,
- c) $(-1)^n n = o(n^2)$ for $n \rightarrow +\infty$,
- d) $2n = o(n^3)$ for $n \rightarrow +\infty$,
- e) $3 \cos(\sqrt{n}) = \mathcal{O}(1)$ for $n \rightarrow +\infty$,
- f) $4n^3 \sin(n) = \mathcal{O}(n^3)$ for $n \rightarrow +\infty$,
- g) $-3n^3 \cos(2n^2 + 3) = o(n^4)$ for $n \rightarrow +\infty$,
- h) $\frac{5 \sin(n)}{n} = \mathcal{O}\left(\frac{1}{n}\right)$ for $n \rightarrow +\infty$,
- i) $\frac{3 \sin(n)}{n^2} = o\left(\frac{1}{n}\right)$ for $n \rightarrow +\infty$.

Recall

$$\begin{aligned} a_n = \mathcal{O}(b_n) &\Leftrightarrow \exists C > 0, K > 0 \text{ such that } |a_n| \leq C|b_n|, \forall n > K, \\ a_n = o(b_n) &\Leftrightarrow \forall C > 0, \exists K > 0 \text{ such that } |a_n| < C|b_n|, \forall n > K. \end{aligned}$$

Solution.

- a) $|n^2 + 2n^4| \leq |n^2| + 2|n^4| \leq 3|n^4|, \forall n > 0$.
- b) $|(-1)^n n| = |(-1)^n| |n| \leq 1 \cdot |n|, \forall n > 0$.
- c) $|(-1)^n n| = |(-1)^n| |n| = 1 \cdot |n| < C|n|^2, \forall C > 0 \text{ and } \forall n > \frac{1}{C}$.

d) $|2n| < C|n^3|, \forall C > 0 \text{ and } \forall n > \sqrt{\frac{2}{C}}.$

e) $|3\cos(\sqrt{n})| < 3, \forall n > 0.$

f)

Note: in the following exercises it can be useful the fact that given sequences (a_n) , (b_n) and (c_n) , for $n \rightarrow +\infty$ it holds true

- $a_n = o(b_n) \Rightarrow a_n = \mathcal{O}(b_n)$ (“little o is stronger than big $\mathcal{O}$ ”)
- $a_n = o(b_n)$ and $b_n = o(c_n) \Rightarrow a_n = o(c_n)$ (transitive property)
- $a_n = \mathcal{O}(b_n)$ and $b_n = \mathcal{O}(c_n) \Rightarrow a_n = \mathcal{O}(c_n)$ (transitive property).

Similar properties hold for function, formulation is good exercise for the reader.

Exercise 2.2: Decide which of the following statements are true and *justify your choice with proof*.

- a) $2\sqrt[4]{n} + 3\sqrt[3]{n} - 4 = \mathcal{O}(\sqrt[3]{n})$ for $n \rightarrow +\infty$,
- b) $2\sqrt[4]{n} + 3\sqrt[3]{n} - 4 = \mathcal{O}(3n)$ for $n \rightarrow +\infty$,
- c) $2\sqrt[4]{n} + 3\sqrt[3]{n} - 4 = \mathcal{O}(\sqrt[4]{n} - 4)$ for $n \rightarrow +\infty$.
- d) $2\sqrt[4]{n} + 3\sqrt[3]{n} - 4 = o(n)$ for $n \rightarrow +\infty$.

Solution. a) true, b) true, c) false, d) true. The proof is good exercises for the reader.

Exercise 2.3: Decide whether the following statements are true or false and *justify your choice with proof*.

- a) $\frac{3}{\sqrt{n}} - 2 = \mathcal{O}(1)$ for $n \rightarrow +\infty$,
- b) $\frac{4}{n} - 3 = \mathcal{O}(1)$ for $n \rightarrow +\infty$,
- c) $\frac{2}{\sqrt{n}} - 1 - \frac{3}{n} = \mathcal{O}(1)$ for $n \rightarrow +\infty$.
- d) $\frac{2}{\sqrt{n}} - \frac{3}{n} = \mathcal{O}(\frac{1}{\sqrt{n}})$ for $n \rightarrow +\infty$.

Solution. a) true, b) true, c) true, d) true. The proof is good exercises for the reader.

Exercise 2.4: Decide whether the following statements are true or false and *justify your choice with proof*.

- a) $3n + 5n^3 = o(n^3)$ for $n \rightarrow +\infty$.
- b) $3n + 5n^3 = \mathcal{O}(n^3)$ for $n \rightarrow +\infty$.
- c) $3n + 5n^3 = o(n)$ for $n \rightarrow +\infty$.
- d) $3n + 5n^3 = \mathcal{O}(n)$ for $n \rightarrow +\infty$.
- e) $3n + 5n^3 = o(n^4)$ for $n \rightarrow +\infty$.

f) $3n + 5n^3 = \mathcal{O}(n^4)$ for $n \rightarrow +\infty$.

Solution. a) false, b) true, c) false, d) false, e) true f) true. The proof is good exercises for the reader.

Exercise 2.5: Decide whether the following statements are true or false and *justify your choice with proof*.

a) $\frac{3}{n} = o\left(\frac{2}{\sqrt{n}}\right)$ for $n \rightarrow +\infty$.

b) $\frac{2}{n} = o\left(\frac{1}{n^2}\right)$ for $n \rightarrow +\infty$.

c) $\frac{3}{n} + \frac{4}{\sqrt{n}} = o\left(\frac{1}{\sqrt{n}}\right)$ for $n \rightarrow +\infty$.

d) $\frac{3}{n} + \frac{4}{\sqrt{n}} = \mathcal{O}\left(\frac{1}{n}\right)$ for $n \rightarrow +\infty$.

Solution. a) true, b) false, c) false, d) false. The proof is good exercises for the reader.

Exercise 2.6: Decide whether the following statements are true or false and *justify your choice with proof*.

a) $3\sqrt{n} + 2\sqrt[3]{n} + 4n^3 = \mathcal{O}(n^3)$ for $n \rightarrow +\infty$.

b) $3\sqrt{n} + 2\sqrt[3]{n} + 4n^3 = \mathcal{O}(n)$ for $n \rightarrow +\infty$.

c) $3\sqrt{n} + 2\sqrt[3]{n} + 4n^3 = \mathcal{O}(\sqrt[3]{n})$ for $n \rightarrow +\infty$.

d) $3\sqrt{n} + 2\sqrt[3]{n} + 4n^3 = o(n^5)$ for $n \rightarrow +\infty$.

e) $3\sqrt{n} + 2\sqrt[3]{n} + 4n^3 = \mathcal{O}(n^5)$ for $n \rightarrow +\infty$.

Solution. a) true, b) false, c) false, d) true e) true. The proof is good exercises for the reader.

2.2 Monotonic sequences

Exercise 2.7: Which of the following sequences are monotonic? Which are eventually monotonic? Specify the type of monotonicity (increasing, strictly increasing, decreasing,..., eventually increasing,...) Which of the following sequences are bounded below, bounded above, bounded (from both sides)? Find the appropriate bounds (not necessarily suprema and infima).

a) $(\sin(2\pi n))_{n=1}^{\infty}$,

b) $(2n - 2)_{n=1}^{\infty}$,

c) $\left(\frac{1}{2n - 5}\right)_{n=1}^{\infty}$,

d) $\left(\left\lfloor \frac{n}{2} \right\rfloor\right)_{n=1}^{\infty}$,

e) $(\operatorname{sgn}(n - 20))_{n=1}^{\infty}$.

Solution. c) $a_n = \left(\frac{1}{2n-5}\right)_{n=1}^{\infty}$.

To study the monotonicity of the sequence we have to see whether the sequence is increasing or decreasing, thus we study the sign of $a_{n+1} - a_n$:

$$a_{n+1} - a_n = \frac{1}{2(n+1)-5} - \frac{1}{2n-5} = \frac{-2}{(2n-3)(2n-5)} \geq 0 \quad ?$$

At numerator we have $-2 < 0$ and at denominator

$$(2n-3)(2n-5) > 0$$

if $n > \frac{5}{2}$ or $n = 1$ (Remember that n is a natural number !) Thus, $a_{n+1} - a_n < 0$ for $n \geq 3$. This implies that the sequence is *eventually* strictly decreasing. By the theorem on the limit of a monotonic sequence, in the case of a decreasing sequence, we know that $\lim_{n \rightarrow +\infty} a_n$ gives information on $\inf_{n \geq 3} a_n$. In this case we have

$$\inf_{n \geq 3} a_n = \lim_{n \rightarrow +\infty} \frac{1}{2n-5} = 0.$$

Is 0 a lower bound for the whole sequence? We can be sure that it is a lower bound only for $\{a_3, a_4, a_5, \dots\}$, since by the monotonicity of the sequence, we know only that

$$a_3 \geq a_4 \geq a_5 \geq \dots \quad (2.1)$$

From formula (2.1) we also see that a_3 is an upper bound for $\{a_3, a_4, a_5, \dots\}$. However is a_3 an upper bound for the whole sequence? We need to check the values of a_1 , a_2 and a_3 . We have

$$a_1 = -\frac{1}{3}, \quad a_2 = -1, \quad a_3 = 1.$$

The smallest one is $a_2 = -1 < 0$, thus a lower bound is -1 . The greatest one is $a_3 = 1$, thus an upper bound is given by $a_3 = 1$.

Exercise 2.8: Which of the following sequences are monotonic? Which are eventually monotonic? Specify the type of monotonicity (increasing, strictly increasing, decreasing, ..., eventually increasing, ...) Which of the following sequences are bounded from below, bounded from above, bounded (from both sides)? Find the appropriate bounds (not necessarily suprema and infima).

1. $a_n = \frac{-2n+3}{n+1},$

2. $a_n = e^n - n,$

3. $a_n = \frac{(n-3)^2}{2^n},$

4. $a_n = 4n - n^2,$

5. $\left(\sqrt{2n-1} - \sqrt{2n}\right)_{n=1}^{\infty},$

6. $\left(\sqrt{2n-1} - \sqrt{3n}\right)_{n=1}^{\infty},$ (analyse only the boundedness)

7. $a_n = n^2 - 2n,$

8. $(n + (-1)^n)_{n=1}^{\infty}$.

Exercise 2.9: Which of the following sequences are monotonic? Which are eventually monotonic? Specify the type of monotonicity (increasing, strictly increasing, decreasing,..., eventually increasing,...) Which of the following sequences are bounded below, bounded above, bounded (from both sides)? Find the appropriate bounds (not necessarily suprema and infima).

a) $(1 - 3 \cos(2\pi n^2))_{n=1}^{\infty}$,

b) $(3^{-n^2+9})_{n=1}^{\infty}$,

c) $\left(\frac{5}{n^2 - 4}\right)_{n=1}^{\infty}$,

d) $(n^3 - 8)_{n=1}^{\infty}$,

e) $\left(\frac{3 + 2n^2}{2 + n^2}\right)_{n=1}^{\infty}$.